CHAPTER 5

BIOTECHNOLOGY: PRINCIPLES AND PROCESSES

CHAPTER ESSENCE AT A GLANCE:
Recombinant DNA engineering relies on restriction endonucleases, cloning vectors, and bioreactor bioprocessing for recombinant protein synthesis.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Restriction Enzymes and Molecular Scissors:** Restriction endonucleases recognize specific palindromic nucleotide sequences in DNA and cleave sugar-phosphate backbones (e.g., EcoRI cleaves between G and A in 5'-GAATTC-3', producing single-stranded overhangs termed sticky ends). DNA ligase joins complementary sticky ends to construct recombinant DNA molecules.

• **Cloning Vector Characteristics: pBR322:** Plasmids used as cloning vectors possess: (1) Origin of replication (ori) controlling autonomous replication and copy number, (2) Selectable markers like antibiotic resistance genes (ampR and tetR) identifying transformants, and (3) Unique restriction sites located within selectable markers for insertional inactivation. Rop codes for proteins involved in plasmid replication.

• **Insertional Inactivation and Selection:** Insertional inactivation differentiates recombinants: inserting a foreign gene into tetR inactivates tetracycline resistance. Blue-white screening provides an alternative: insertional inactivation of the beta-galactosidase gene (lacZ); non-recombinants produce active enzyme forming blue colonies in the presence of chromogenic X-gal, while recombinants produce white colonies.

• **Polymerase Chain Reaction and Bioreactor Cultivation:** PCR amplifies target DNA across three thermal cycles: Denaturation (94°C separating strands), Annealing (54°C hybridizing oligonucleotide primers), and Extension (72°C using thermostable Taq polymerase from Thermus aquaticus; 30 cycles amplify DNA a billion-fold). Bioreactors provide optimal temperature, pH, oxygen, and nutrients for mass scale culture, followed by downstream separation and purification.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$5'\text{-G}\downarrow\text{AATTC-3'} \Longleftrightarrow 3'\text{-CTTAA}\uparrow\text{G-5'} \quad [\text{EcoRI Palindromic Cleavage Sequence}]$$

Specific Restriction Cleavage Producing Complementary Sticky Ends

$$\text{PCR Cycles} = \text{Denaturation } (94^\circ\text{C}) \rightarrow \text{Annealing } (54^\circ\text{C}) \rightarrow \text{Extension } (72^\circ\text{C: Taq Pol})$$

Three-Step Thermal Amplification Cycle in PCR

$$\text{PCR Amplification} = N_0 \times 2^n \quad [n = 30 \text{ cycles} \rightarrow \sim 10^9\text{-fold DNA copies}]$$

Exponential Amplification Yield of Target DNA Duplexes

$$\text{pBR322 Architecture} = \text{ori (copy control)} + \text{amp}^R \text{ and tet}^R \text{ (selectable markers)} + \text{rop}$$

Standard Artificial Plasmid Vector Layout

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Taq polymerase is isolated from the thermophilic bacterium Thermus aquaticus because it remains enzymatically active at the high temperature of denaturation (94°C).

• **Alert:** In blue-white selection, recombinant bacterial colonies are WHITE (due to insertional inactivation of the lacZ gene), while non-recombinant colonies are BLUE.

• **Alert:** Chilled ethanol is specifically used to precipitate purified DNA, which can then be spooled out as fine threads.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What are palindromic nucleotide sequences in DNA? Illustrate with the specific restriction site recognized and cleaved by EcoRI. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• Definition: A palindromic sequence in DNA is a sequence of base pairs that reads the same on both strands when reading direction is kept the same in 5' to 3' orientation. **[1 Mark]**

• EcoRI palindrome: Cleaves between G and A on both strands: 5'-G A A T T C-3' and 3'-C T T A A G-5', generating staggered single-stranded sticky ends. **[1 Mark]**

2. Why is Taq polymerase preferred over normal DNA polymerase in the Polymerase Chain Reaction (PCR)? Name the organism from which it is isolated. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• Thermal stability: Taq polymerase is isolated from the thermophilic bacterium Thermus aquaticus. **[1 Mark]**

• PCR advantage: It remains enzymatically active and stable during the high-temperature denaturation step (94°C), eliminating the need to add fresh enzyme after each thermal cycle. **[1 Mark]**

3. Explain the principle of insertional inactivation using the beta-galactosidase (lacZ) chromogenic assay for distinguishing recombinant from non-recombinant bacterial colonies. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• A recombinant DNA is inserted within the coding sequence of the beta-galactosidase (lacZ) enzyme gene in a plasmid vector. **[1 Mark]**

• This insertion disrupts the coding frame, causing insertional inactivation of the enzyme so that no functional beta-galactosidase is produced. **[1 Mark]**

• When grown on medium containing chromogenic substrate (X-gal), non-recombinant colonies produce active enzyme turning blue, whereas recombinant colonies with disrupted genes produce white colonies. **[1 Mark]**

4. Describe the three sequential thermal steps of the Polymerase Chain Reaction (PCR) and calculate the theoretical number of DNA duplexes produced after 30 cycles. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Denaturation: Double-stranded target DNA is heated to high temperature (about 94°C) to break hydrogen bonds and separate into single strands. **[0.75 Mark]**

• Annealing: Reaction temperature is lowered to around 50 to 56°C, allowing two oligonucleotide primers to hybridize to complementary sequences on each single-stranded DNA template. **[0.75 Mark]**

• Extension: Temperature is raised to 72°C; thermostable Taq polymerase extends primers in 5' to 3' direction using free deoxynucleotides. **[0.75 Mark]**

• Amplification yield: 30 cycles amplify target DNA approximately 2^{30} times, yielding about 1 billion copies. **[0.75 Mark]**

5. Draw a diagram of cloning vector pBR322 and explain the functional significance of the origin of replication (ori) and antibiotic resistance genes (ampR and tetR). **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Origin of replication (ori): Specific DNA sequence where replication starts; any foreign DNA linked to it replicates inside host cells, and it controls the copy number of the linked DNA. **[1 Mark]**

• Selectable markers (ampR and tetR): Antibiotic resistance genes that allow selection of transformed cells; cells taking up pBR322 grow on ampicillin and tetracycline plates, while non-transformants die. **[1 Mark]**

• Cloning sites: Unique restriction sites located inside ampR (PstI, PvuI) and tetR (BamHI, SalI) enable insertional inactivation for recombinant screening. **[1 Mark]**

6. Compare microinjection with biolistics (gene gun) as vectorless methods for introducing recombinant DNA into host cells. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Microinjection: Recombinant DNA is directly injected into the nucleus of an animal cell using a microscopic glass micropipette under a micromanipulator. **[1.5 Marks]**

• Biolistics (gene gun): Suitable for plant cells; cells are bombarded with high-velocity microscopic gold or tungsten microparticles coated with recombinant DNA to penetrate rigid cell walls. **[1.5 Marks]**

7. Describe the structural features of a stirred-tank bioreactor and explain why it is superior to a simple laboratory shake flask for the mass culture of recombinant microorganisms. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Structural features: A large cylindrical vessel with a curved base; fitted with a motorized stirrer, an oxygen delivery sparger system, an agitator, a foam breaker, a temperature control jacket, and pH sensors with sampling ports. **[1.5 Marks]**

• Superiority over shake flasks: Laboratory shake flasks cannot handle large commercial volumes (100 to 1000 liters) and lack continuous parameter monitoring. **[1 Mark]**

• Optimal biological conditions: The bioreactor continuously maintains optimal temperature, dissolved oxygen saturation, pH, substrate concentration, and salts, maintaining cells in their exponential log phase of growth for maximum recombinant protein yield. **[1.5 Marks]**